

ALEXANDER DORNBACK

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MS, Robotics @ Columbia | Controls, Hardware, Sim2Real | Available Summer 2026

EDUCATION

Columbia University MS, Mechanical Engineering , 3.95/4 Robotics, Automation, and Controls Concentration <i>Research Affiliation: Creative Machines Laboratory, Prof. Hod Lipson (January 2026 – Present)</i> Relevant Courses: Robot Learning, Robotics Studio, Discrete Controls, Mechatronics	New York, NY Expected Dec 2026
Case Western Reserve University BS, Biomedical Engineering Biomechanics Concentration	Cleveland, OH May 2022

ENGINEERING PROJECTS

FIDO: an open-source, trainable, 3D-printed quadruped robot

- Implemented random search/hill-climbing for gait parameter optimization and PPO for sensorless joint neutral estimation
- Achieved stance in 10 minutes of parallel-environment PPO training on 4 custom linkage-drive legs in MuJoCo
- Achieved stable walking in 30-minute training and completed sim-to-real pipeline for gait of 25 cm/s on real-world deployment

Hybrid MPC-CBF impact shaping for bipedal locomotion

- Selected CBF after PID and vanilla MPC exceeded 3.34N actuator limits on impact; enforced safety across drops up to 40 cm
- Built kinematic TTC vision estimator as feedforward input to dynamically adjust prediction horizon pre-impact
- At 40cm, MPC-CBF achieved 6.71N GRF vs. 8.00N for PID – a 16% reduction while ensuring hardware limit adherence

Typhoon: A Polar IDEX System for Simultaneous Multi-Material 3D Printing

- Designed and built a novel polar 3D printer in Onshape on a Creality Ender chassis with dual active hotends on a single rail
- Demonstrated simultaneous dual-material deposition with zero-purge waste, validating the core polar IDEX architecture
- Simulink dynamic analysis for polar axis stepper selection revealed 2-2.5x resonance amplification missed in static motor sizing

Tendon-driven robotic bending actuator for semi-autonomous intubation

- Designed a superelastic nitinol tendon-driven actuator with scotch-yoke transmission, achieving 30°/s tip deflection
- Autonomous IR closed-loop tip control achieved 90% accuracy and first-pass intubation in 20-second clinical benchmark

Drone-based system for autonomous building maintenance

- Built a tether-powered drone with continuous power delivery and modular pressure-washing tool head field-tested at 20 feet
- Designed tandem tethered power and water delivery systems to account for pressure drop and wire resistive loss

WORK EXPERIENCE

Weill Cornell Medicine Dario Institute Robotics Engineer	New York, NY Jul 2022 - Sep 2025
• Led 3 engineers over 12 months to deliver a clinically validated intubation robot tested by untrained and ENT/EMS operators • Designed and fabricated 3 mechanical testbeds tracking twist, stretch, and bend of a silicone capacitive force sensor • Built CFD-validated multi-material segmented printing workflow for 65 CT angiograms to assess coronary artery disease risk	
Haima Therapeutics Formulation Scientist	Cleveland, OH Oct 2020 - Jun 2022
• Characterized over 200 nanoparticle batches, raising \$3m in DARPA grant funding alongside C-suite and scientist team • Co-author: Sekhon et al. Wiley Adv. Science, 2025 (https://doi.org/10.1002/adv.202600002)	

SKILLS

- Controls and Autonomy: Model Predictive Control, Control Barrier Functions, PID, RL (PPO), Sim2Real, Trajectory Optimization
- Simulation and Modeling: MuJoCo, Kinematics/Dynamics, Impact Shaping, CFD
- Mechanics and Mechatronics: SolidWorks (CSWA), Onshape, GD&T, Actuator/Transmission Design, Silicone Casting, 3D Printing
- Software and Tooling: Python, Computer Vision, MATLAB, Git, Arduino, Raspberry Pi, LabVIEW